

# Bitcoin Residual Dynamics Playground

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$$\dot{y}_1 = y_2 - Z_A y_2 z + Z_{\text{mix}} (y_1 - y_2) \quad (1)$$

$$\dot{y}_2 = \underbrace{\alpha y_2 |y_2|^{M-1} - \gamma y_1 |y_1|^{N-1}}_{\text{}} + Z_B y_1 z - \frac{\kappa \operatorname{sgn}[N(t)] + y_2}{t} + \frac{\sigma(t)}{t} \dot{W}_t \quad (2)$$

$$\dot{z} = -Z_C z + Z_D y_1 + Z_E y_1 y_2 \quad (3)$$

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## 1. LPPL Oscillator (Core)

$$\dot{y}_1 = y_2 \quad (4)$$

$$\dot{y}_2 = \alpha y_2 |y_2|^{M-1} - \gamma y_1 |y_1|^{N-1} \quad (5)$$

## 2. Diffusion Extension

$$\dot{y}_2 = -\frac{\kappa \operatorname{sgn}[N(t)] + y_2}{t} + \frac{\sigma(t)}{t} \dot{W}_t \quad (6)$$

$$N(t) = \nu(t) - n(t), \quad \nu(t) := n_{PL} + t y_2 \quad (7)$$

## 3. Wang-like 2-Scroll Extension

$$\dot{y}_1 = y_2 - Z_A y_2 z + Z_{\text{mix}} (y_1 - y_2) \quad (8)$$

$$\dot{y}_2 = \alpha y_2 |y_2|^{M-1} - \gamma y_1 |y_1|^{N-1} + Z_B y_1 z \quad (9)$$

$$\dot{z} = -Z_C z + Z_D y_1 + Z_E y_1 y_2 \quad (10)$$

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Original Wang Multi-Scroll:

$$\dot{y}_1 = Z_{\text{mix}} (y_1 - y_2) - y_2 z \quad (11)$$

$$\dot{y}_2 = -b y_2 + y_1 z \quad (12)$$

$$\dot{z} = -Z_C z + Z_D y_1 + y_1 y_2 \quad (13)$$

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## LPPL Oscillator

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| $y_1$                  | price residual in log-space (displacement from power-law) |
| $y_2 = \dot{y}_1$      | residual change rate (velocity)                           |
| $\alpha, \gamma, M, N$ | LPPL parameters                                           |

## Laplace Diffusion

|             |                                                         |
|-------------|---------------------------------------------------------|
| $n_{PL}$    | Bitcoin power-law mean exponent                         |
| $\nu(t)$    | local drifting exponent: $n_{PL} + t \dot{y}_1$         |
| $n(t)$      | real current Bitcoin exponent, cycle-dependent          |
| $N(t)$      | exponent deviation from long term mean: $\nu(t) - n(t)$ |
| $\kappa$    | restoring force (sign-based)                            |
| $\sigma(t)$ | noise diffusion coefficient (controls Laplace width)    |
| $\dot{W}_t$ | Gaussian white noise increment                          |

## Wang 2-Scroll

|                  |                       |
|------------------|-----------------------|
| $z$              | feedback memory       |
| $Z_A, Z_B$       | active coupling gains |
| $Z_C, Z_D, Z_E$  | feedback dynamics     |
| $Z_{\text{mix}}$ | mixing term           |

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## Parameter Values

|                                     |
|-------------------------------------|
| $\alpha = -7.4 \times 10^{-4}$      |
| $\gamma = 3.0 \times 10^{-3}$       |
| $M = 1.071$                         |
| $N = 3.0$                           |
| $n_{PL} = 5.6574$                   |
| $\kappa = 108.62$                   |
| $\sigma = 40\text{--}90$            |
| $Z_A, Z_B = 8 \times 10^{-3}$       |
| $Z_C = 3.9 \times 10^{-3}$          |
| $Z_D = 10^{-6}$                     |
| $Z_E = 2.0$                         |
| $Z_{\text{mix}} = 2 \times 10^{-4}$ |